João Vitor Cavalcanti da Silva

Senior Software Engineer

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SUMMARY

Senior software engineer. Since 2023, shipping production systems for Brazilian public healthcare: the product, the data pipelines and the Linux servers they run on. TypeScript, Node.js, React, Next.js and PostgreSQL. The reporting platform is in daily use across 400+ municipalities, with 10k+ monthly active users.

EXPERIENCE

Founder & Senior Software Engineer · Apr 2026 — present

Quiral Labs · Self-employed · Recife, Brazil · Remote

Engineering studio. Infrastructure and backend contracts for software companies.

- Cut a healthcare client's hosting cost by about US$72k a year, moving 300+ municipal servers onto VPSs about 10x cheaper per machine, with all traffic through WireGuard tunnels.
- Built the control plane for that fleet: inventory, Ansible provisioning (firewall, NAT, WireGuard, nginx) and a health agent on 300+ machines, plus monitors that restore a dropped tunnel within seconds.
- Built automated backups of every municipal database to S3-compatible storage, with multipart uploads and a 3-day / 3-month retention policy.
- Doubled the speed of the nightly sync that loads millions of rows per municipality from 300+ remote databases into one PostgreSQL warehouse, using Node.js streams and Bull/Redis queues.

Senior Software Engineer · Apr 2023 — present

Wi Consultoria · Contract · Recife, Brazil · Remote

Management software for municipal health departments in Brazil's public health system.

- Built the primary-care reporting platform used in 400+ municipalities (10k+ monthly active users): one tenant per city, PDF and CSV exports, and real-time patient call screens. Next.js, React, Express, Knex, PostgreSQL, Socket.IO.
- Shipped WhatsApp campaigns on Bull/Redis queues and an engagement dashboard of feature usage, cohorts and deep-link conversion.
- Grew organic search clicks 47x (+4,648%) and impressions 7x (+592%) in three months on the public site, with server-side rendering, JSON-LD and a sitemap.
- Kept production up through deploys: nginx canary across two app servers (PM2), leader-locked workers, and jobs that recover stuck syncs.
- Set up the engineering workflow across 15+ repositories so changes from coding agents land only after the tests pass.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, SQL, Bash

Interface: React, Next.js, Tailwind CSS

Server: Node.js, Express, Fastify, Knex, Bull, REST, WebSocket

Data: PostgreSQL, Redis

Infra: Linux, Nginx, PM2, Ansible, WireGuard, Amazon Web Services (AWS) S3, Git, CI/CD

EDUCATION

Computer Science · Jan 2025 — present

UniFavip Wyden

Bachelor's degree in Computer Science, in progress.

CERTIFICATIONS

2025 Cloud Architecting — AWS Academy

2025 Cloud Foundations — AWS Academy

2025 Cyber Threat Management — Cisco

2025 1st place (team) — AWS Hackathon Nordeste

LANGUAGES

Portuguese — Native

English — Intermediate